Practice Set A

Class 12 Mathematics, Chapter 6: Application of Derivatives

Maximum marks: **52** **20** questions

Name _____ Date _____ Score _____ / 52

1. All questions are compulsory. Marks are shown against each question.
2. Show every step of working: steps carry marks, not just the final answer.
3. Where a question offers OR, attempt only one of the two choices.

Q No.	Question	Marks

Level 1

8 marks

1 MCQ, Board-pattern — 1 mark

The radius of a circular plate is increasing at the rate of 0.2 cm/s. The rate at which its area is increasing when the radius is 5 cm is:

(A) π cm^2/s **(B)** 2π cm^2/s **(C)** 10π cm^2/s **(D)** 25π cm^2/s

2 MCQ, CBSE SQP 2024-25 — 1 mark

The largest interval in which the function f defined by $f(x) = e^x$ is strictly increasing, is:

(A) $[1, \infty)$ **(B)** $(-\infty, 0)$ **(C)** $(-\infty, \infty)$ **(D)** $(0, \infty)$

3 MCQ, CBSE SQP 2025-26 — 1 mark

A function $f(x) = 10 - x - 2x^2$ is increasing on the interval:

(A) $\left(-\infty, -\frac{1}{4}\right]$ **(B)** $\left(-\infty, \frac{1}{4}\right)$ **(C)** $\left[-\frac{1}{4}, \infty\right)$ **(D)** $\left[-\frac{1}{4}, \frac{1}{4}\right]$

4 MCQ, CBSE 2025 — 1 mark

The absolute maximum value of the function $f(x) = x^3 - 3x + 2$ in $[0, 2]$ is:

(A) 0 **(B)** 2 **(C)** 4 **(D)** 5

5 MCQ, Board-pattern — 1 mark

The local minimum value of $f(x) = x^2 + \frac{16}{x}$, $x > 0$, is:

(A) 2 **(B)** 8 **(C)** 12 **(D)** 16

Q No.	Question	Marks
6	Assertion-Reason, CBSE SQP 2023-24 Let $f(x)$ be a polynomial function of degree 6 such that $\frac{d}{dx}\big(f(x)\big) = (x-1)^3(x-3)^2$. **Assertion (A):** $f(x)$ has a minimum at $x = 1$. **Reason (R):** When $\frac{d}{dx}\big(f(x)\big) < 0$ for all $x \in (a-h, a)$ and $\frac{d}{dx}\big(f(x)\big) > 0$ for all $x \in (a, a+h)$, where h is an infinitesimally small positive quantity, then $f(x)$ has a minimum at $x = a$, provided $f(x)$ is continuous at $x = a$. **(A)** Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A). **(B)** Both Assertion (A) and Reason (R) are true, but Reason (R) is **not** the correct explanation of the Assertion (A). **(C)** Assertion (A) is true, but Reason (R) is false. **(D)** Assertion (A) is false, but Reason (R) is true.	1 mark
7	Short answer, CBSE 2025 The side of an equilateral triangle is increasing at the rate of 3 cm/s. At what rate is its area increasing when the side of the triangle is 15 cm?	2 marks

Level 2

19 marks

8	Short answer, CBSE 2025 Find the intervals in which the function $f(x) = 5x^3 - 3x^2$ is (i) increasing (ii) decreasing.	3 marks
9	Short answer, CBSE SQP 2025-26 A spherical ball of ice melts in such a way that the rate at which its volume decreases at any instant is directly proportional to its surface area. Prove that the radius of the ice ball decreases at a constant rate.	2 marks
10	Short answer, PYQ-style Find the local maximum and local minimum values of $f(x) = 3x^4 + 4x^3 - 12x^2 + 12$.	3 marks
11	Short answer, CBSE SQP 2023-24 Find the interval/s in which the function $f : \mathbb{R} \to \mathbb{R}$ defined by $f(x) = xe^x$ is increasing.	3 marks
12	Short answer, CBSE SQP 2024-25 A kite is flying at a height of 3 metres and 5 metres of string is out. If the kite is moving away horizontally at the rate of 200 cm/s, find the rate at which the string is being released.	3 marks
13	Short answer, CBSE 2025 Find the absolute maximum and absolute minimum of the function $f(x) = 2x^3 - 15x^2 + 36x + 1$ on $[1, 5]$.	3 marks
14	Short answer, CBSE SQP 2023-24 Find the maximum profit that a company can make, if the profit function is given by $P(x) = 72 + 42x - x^2$, where x is the number of units and P is the profit in rupees.	2 marks

Q No.	Question	Marks

Level 3 — 25 marks

15 Long answer, PYQ-style — 5 marks

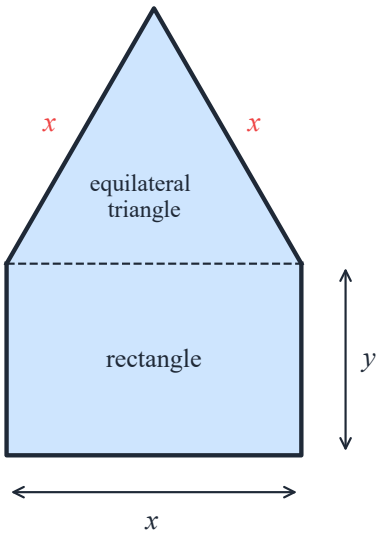
An open tank with a square base and vertical sides is to be constructed from a metal sheet so as to hold a given quantity of water. Show that the cost of the material will be least when the depth of the tank is half of its width.

16 Long answer, PYQ-style — 5 marks

Show that the altitude of the right circular cone of maximum volume that can be inscribed in a sphere of radius r is $\frac{4r}{3}$. Also show that the maximum volume of the cone is $\frac{8}{27}$ of the volume of the sphere.

17 Long answer, PYQ-style — 5 marks

A window is in the form of a rectangle surmounted by an equilateral triangle. The perimeter of the window is 12 m. Find the dimensions of the rectangle that will produce the largest area of the window.



18 Short answer, PYQ-style — 3 marks

Find the intervals in which the function $f(x) = \sin^4 x + \cos^4 x$, $0 \le x \le \frac{\pi}{2}$, is strictly increasing or strictly decreasing.

19 Short answer, PYQ-style — 3 marks

A water tank has the shape of an inverted right circular cone with its axis vertical and vertex lowermost. Its semi-vertical angle is $\tan^{-1}(0.5)$. Water is poured into it at a constant rate of $5\ \mathrm{m}^3/\mathrm{h}$. Find the rate at which the level of the water is rising at the instant when the depth of water in the tank is 4 m.

Q No.	Question	Marks
20	Case study, CBSE SQP 2025-26 **Case Study:** LED bulbs are energy-efficient because they use much less electricity than traditional bulbs while producing the same amount of light. A company manufactures a new type of energy-efficient LED bulb. The cost of production and the revenue generated by selling x bulbs (in an hour) are modelled as $C(x) = 0.5x^2 - 10x + 150$ and $R(x) = -0.3x^2 + 20x$ respectively, where $C(x)$ and $R(x)$ are both in ₹. Use the given models to answer the following questions. **(i)** Derive the profit function $P(x)$. — 1 mark **(ii)** Find the critical points of $P(x)$. — 1 mark **(iii)(a)** Determine whether the critical points correspond to a maximum or a minimum profit by using the second derivative test. — 2 marks OR **(iii)(b)** Identify the possible practical value of x (i.e. the number of bulbs that can realistically be produced and sold) that can maximise the profit, if the resources available and the expenditure on machines allow the company to produce a minimum of 10 but not more than 18 bulbs per hour. Also calculate the maximum profit. — 2 marks	4 marks

Answer key

Chapter 6: Application of Derivatives, Practice Set A. Full step-wise solutions are in the solutions sheet.

Level 1

1	(B) $2\pi \text{ cm}^2/\text{s}$
2	(C) $(-\infty, \infty)$
3	(A) $\left(-\infty, -\frac{1}{4}\right]$
4	(C) 4
5	(C) 12
6	(A) Both true; R is the correct explanation of A.
7	Area increases at $\frac{45\sqrt{3}}{2} \text{ cm}^2/\text{s}$

Level 2

8	(i) Increasing in $(-\infty, 0] \cup \left[\frac{2}{5}, \infty\right)$ (ii) Decreasing in $\left[0, \frac{2}{5}\right]$
9	$\frac{dr}{dt} = -k$, a constant; so the radius decreases at a constant rate.
10	Local maximum value 12 (at $x = 0$); local minimum values -20 (at $x = -2$) and 7 (at $x = 1$)
11	f is increasing in $[-1, \infty)$
12	String is released at 1.6 m/s ($= 160$ cm/s)
13	Absolute maximum $= 56$ at $x = 5$; absolute minimum $= 24$ at $x = 1$
14	Maximum profit $= ₹\,513$ (at $x = 21$ units)

Level 3

15	Sheet area (hence cost) is least when $h = \frac{x}{2}$, i.e. depth $=$ half the width
16	Altitude $= \frac{4r}{3}$; maximum volume $= \frac{32\pi r^3}{81} = \frac{8}{27} \times \frac{4}{3}\pi r^3$
17	Width $= \frac{12}{6-\sqrt{3}} = \frac{4(6+\sqrt{3})}{11}$ m (≈ 2.81 m), height $= \frac{18-6\sqrt{3}}{6-\sqrt{3}} = \frac{30-6\sqrt{3}}{11}$ m (≈ 1.78 m)
18	Strictly decreasing in $\left(0, \frac{\pi}{4}\right)$; strictly increasing in $\left(\frac{\pi}{4}, \frac{\pi}{2}\right)$
19	The water level rises at $\frac{5}{4\pi}$ m/h (≈ 0.40 m/h)
20	(i) $P(x) = -0.8x^2 + 30x - 150$ (ii) $x = 18.75$ (iii)(a) $P''(x) = -1.6 < 0$, so maximum profit (iii)(b) $x = 18$ bulbs, maximum profit $= ₹\,130.80$